



Power analysis as a tool to analyse trade-offs between ecosystem services in forest management: A case study from nine European countries

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ABSTRACT

Forests are of major importance to people, providing fundamental ecosystem services (ESs). Increasing the supply of an ES might negatively affect the supply of another ES. For example, increasing game densities might reduce timber production. Such trade-offs among ESs may lead to conflicts between actors interested in prioritizing different ESs. This study describes which actors dominated conflicts about ES trade-offs, and which power strategies they used to do so. Forest management practices and resulting trade-offs between ESs differ widely among the studied countries: Germany, Ireland, Italy, Lithuania, Portugal, Slovakia, Sweden, the Netherlands, and Turkey. We triangulated 220 qualitative interviews, literature review, document analysis, and participatory observations. We mapped the interests of actors in ESs and identified conflicts between interests. We tested three hypotheses about which actors were more or less powerful, enabling them to be winners and losers in ES conflicts. Cultural and regulating and maintenance ESs played an important role in conflicts about forest ES trade-offs. We identified the power relations of actors with different interests in ES. Local interests often dominated national interests. Actors interested in provisioning ESs had strong power resources but because of specific biogeophysical, political or economic conditions, actors with interest in regulating and maintenance ES or cultural ESs can have equal or stronger power resources. The study highlights the relevance of including power analysis in ES research.

1. Introduction

Forests are of major importance to people, providing fundamental

ecosystem services (ESs). Forest management practices affect the bundles of ESs generated by forests. The generation of some forest ESs can come at the expense of other ESs (Fares et al., 2015; Felton et al., 2016;

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Borges et al., 2017). Human intervention and/or ecological conditions may prioritize the provision of only some specific ESs, to the detriment of other ones (Van der Plas et al., 2016). As different actors derive wellbeing from different (combinations of) ESs, “winners and losers are created as ES change and trade-offs between different ES can also lead to trade-offs in the wellbeing of different groups of people” (Howe et al., 2014, p. 264). These trade-offs can even cause conflicts among actors interested in different ESs. As actors have different access to ESs and different capacities to manage for particular ESs, the role of power in ES trade-offs is a topic that should not be underestimated (Howe et al., 2014). The power relations of actors determine to a large extent decision-making about ES trade-offs in forest management (Felipe-Lucia et al., 2015; Berbés-Blázquez et al., 2016).

It is, therefore, surprising that attention for conflicts and power relations of actors involved in forest management has been limited thus far in the literature. Many studies have found that trade-offs between ESs are common (Howe et al., 2014). Hicks et al. (2013) also found that different actor groups report proportionally more trade-offs than synergies. Additionally, model-based figures estimate that in the future trade-offs between ESs will increase globally (Alcamo et al., 2005), possibly leading to more conflict situations. Aside from a few studies (Corbera et al., 2007; Howe et al., 2014; Felipe-Lucia et al., 2015; Berbés-Blázquez et al., 2016; Vallet et al., 2019; Felton et al., 2020), the ES concept and research about conflicts and power relations in forest management have been rather disconnected.

Many published ES studies focus on only one or few ESs. This focus does not allow an understanding about trade-offs in ESs and underlying conflicts in general. There is a knowledge gap about the conflicts and underlying power relations of actors taking a broad range of forest ES into account. Considering this broad range allows an understanding of the trade-offs in ESs and the underlying preferences and conflicts (e.g., Zhang et al., 2007; Howe et al., 2014; Borges et al., 2017). Our study focuses on a broad range of forest ESs, including economical, ecological and socio-cultural values. Many studies describe conflicts among actors but there is limited knowledge regarding which actors dominate conflicts about ES trade-offs and with which power strategies they can dominate these conflicts. Our study aimed at increasing the knowledge about the role of power in conflicts due to ES trade-offs. Studying these three different aspects in combination allows increased understanding about how power relations among actors impact forest management conflicts, and as a result, the provision of ESs. Such an understanding is needed to develop successful ES management policies (Felipe-Lucia et al., 2015; Berbés-Blázquez et al., 2016).

Few papers on forest management conflicts adopt an international perspective; systematically collecting, comparing, and presenting forest conflict analysis from several countries (e.g. Gritten et al., 2013; Hellström, 2001). Inclusion of a large number of countries necessitates concise analysis, but the resulting lack of detail is compensated by the insights into general mechanisms of forest management conflicts. Our study benefited from involving multiple cases and an international perspective by including nine different countries.

The following four questions guided our study:

- Who are the actors interested in the various ESs provided by forests in the targeted countries?
- What are the conflicts in regard to trade-offs among forest ESs in the targeted countries?
- Which power relations between these actors characterize forest management in the targeted countries?
- Which actors dominate conflicts about ES trade-offs?
- What relevant similarities and differences exist among targeted countries?

2. Theoretical background

2.1. Ecosystem services

The Millennium Ecosystem Assessment (MA) contributed much to putting ESs on the policy agenda and in attracting political support for conservation (Gómez-Baggethun et al., 2010). Based on the MA study (MA, 2005, p. 40) ESs are defined as the “benefits people obtain from ecosystems” and classified into four categories: (1) provisioning, (2) regulating, (3) cultural and (4) supporting services. Provisioning services are the products obtained from ecosystems, for example wood, fuel, food, or natural medicines. Regulating services are the benefits obtained from the regulation of ecosystem processes, such as climate and air quality regulation, pollination, or pest regulation. Cultural services are the benefits people obtain from ecosystems through spiritual enrichment, recreation, and aesthetic experiences. Supporting services are necessary for the production of all other ecosystem services (MA, 2005). After the publication of the MA, other classifications of ecosystem services gained relevance within the scientific discourse. The TEEB (2010) classification did not contain the category of supporting ESs but included instead habitat services. The Common International Classification of Ecosystem Services (CICES) has been developed by the European Environmental Agency and is also applied by the United Nations (e.g. by the Intergovernmental Panel on Biodiversity and Ecosystem Services (IPBES) and the United Nations Statistical Division (UNSD)). The CICES classification includes (1) provisioning services, (2) regulation and maintenance services, and (3) cultural services. In contrast to the MA classification, biodiversity underlies the supply of many services but it is only directly embedded in the ES “Lifecycle maintenance, habitat and gene pool protection” within the regulation and maintenance services of CICES. The IPBES conceptual framework has been enhanced by the concept of “nature’s contributions to people” (NCP) (Díaz et al., 2018). The NCP concept acknowledges the central role of culture in linking people and nature. Furthermore, it highlights the importance of indigenous and local knowledge.

2.2. Forest ecosystem services trade-offs and conflicts about ecosystem services

Although the list of ESs provides the impression that ESs can be combined in the management of forests, often it is impossible to reach all the ESs simultaneously at a level that is desired by the actors involved (Turkelboom et al., 2018; Van der Plas et al., 2016). As a result, the concept of “trade-off” is very common in the forest ES literature. Turkelboom et al. (2018, p. 567) observed that the term trade-off is mostly used in the sense of “a negative correlation between spatial (or temporal) co-occurrences of ES supplies.” Going back to its original meaning from the economic sciences, trade-offs occur due to the fact that “resources are scarce (and) making decisions requires trading off one goal against another” (Mankiw, 1997, p.3). This is even more relevant when considering that, in economic terms, most of the ESs are public goods or even common goods. This description reflects three key elements that we also consider to be the basis for our study (Turkelboom et al., 2018):

- natural resources are limited,
- humans have to make choices about how to use these resources and if the resources should be used,
- which leads to trade-offs in the use of natural resources and among the actors able to use them.

ES trade-offs in forest management often reflect rivalry among the actors involved. When an ES can only be supplied to a low extent, this may directly affect the wellbeing of one or more groups of actors (Iniesta-Arandia et al., 2014). Hence, if a certain ES is provided at a higher level at the expense of another ES, actors have different interests and values regarding ESs, leading to conflicts (Bonsu et al., 2019).

Powerful actors that dominate forest management can widely realize their interests, making them “winners” of conflicts. Less powerful actors, which cannot realize their interests, are “losers” in these forest management conflicts. Conflicts in this study are defined by the existence of conflicting interests in forest ESs. This understanding of conflict includes (1) different priorities about ES trade-offs among actors and (2) interest of different actors in the same ES, which can cause rivalry. The analysis of forest management conflicts is the core of forest policy research because of its high societal relevance (Eckerberg and Sandström, 2013). Conflicts can be more or less prominent in the political process and practical forest management. Conflicts can be latent or escalate to different degrees if active confrontation between actors takes place.

Numerous studies have examined forest management conflicts in a single case study in varying local settings (e.g. Acosta and Corral, 2015; Juerges et al., 2018; Marques et al., 2020). These studies examine the actors, their interests, and forest management conflicts at a local or regional scale. Often, these type of studies focus on conflict management, for example by implementing participatory management. However, there is a lack of studies including multi-country analysis. Howe et al. (2014) showed the value of multi-country studies by analyzing 231 actual or potential synergies and trade-offs between ESs around the world. In the majority of ES trade-offs, the actors that dominated the conflict acted at the local scale, utilized provisioning services and had a private interest in these services. The evidence for the role of the local scale is less clear. Firstly, actors with private interests often act at the local scale; thus acting at the local scale might not be the actual reason why actors win conflicts about ES trade-offs. Secondly, actors that lost conflicts about ES trade-offs often act at a local scale because many cases considered by Howe et al. (2014) were observed at a local scale. More research is needed about the relevance of scale in ES trade-off conflicts.

The research by Howe et al. (2014) suggests that key indicators exist that are highly relevant for the analysis of ES trade-offs:

- one or more stakeholders have a private interest in benefits that can be derived from a particular ES,
- competition exists between provisioning and non-provisioning ES,
- one or more stakeholders act at the local scale.

What Howe et al. (2014), however, did not include in their analysis was the role of power. The actual choices and outcomes are to a larger extent based on power relations. Because of power asymmetries among ES beneficiaries, actors are asymmetrically affected by ES trade-offs (Felipe-Lucia et al., 2015).

2.3. The concept of power

The concept of power is highly contested within political sciences. Often, four concepts and two dualities of power are distinguished in the literature: ‘power over’ and ‘power to’ (Clegg, 1989; Haugaard, 2002; Barnett and Duvall, 2005; Pansardi, 2012) and agent-based and structural power (Giddens, 1984; Barnett and Duvall, 2005). The term ‘power over’ describes power as the relation between actors (Dahl, 1957). In contrast, the term ‘power to’ refers to the ability of actors to achieve things by themselves, either individually or collectively (Arendt, 1970). Agent-based power particularly looks at individual or collective agencies and their capacity to make a difference in the world (Giddens, 1984), while structural power engages with how systems, institutions or discourses shape or affect human agencies’ thoughts, speech acts, doings and practices (Clegg, 1989; Hunter, 1953). Inspired by Dahl (1957) and Krott et al. (2014, p. 35) define power as “the capability of an actor to influence other actors”, understanding power as ‘power over’ and focusing on an agent-based approach. This understanding of power is used to evaluate the power of actors in the actor-centered power (ACP) approach that was developed in the context of forest policy analysis Krott et al. (2014). The ACP approach is a framework that can be used for exploring power relations among actors. We apply the ACP approach

in this study for exploring power relations of actors involved in conflicts about ESs trade-offs in forest management. The assessment of power in the ACP approach is based on the capability of actors to influence other actors by applying the power strategies of “coercion”, “incentives”, and “dominant information” (Table 1). The ability of an actor to apply these three strategies is denoted as the power resources of the actor. A “winner” of ES conflicts has strong power resources, allowing them to dominate forest management by realizing their interests in ESs. In contrast, a “loser” of ES conflicts has only low power resources, compared to other actors with conflicting interest, making them fully or partially unable to realize their interests.

In natural resource management analysis, the exploration of power relations, forms an important bridge to overcome the gap between the provision of ESs and the well-being of actors involved in forest management.

2.4. Hypotheses about which actors dominate conflicts about ES trade-offs

Many studies describe conflicts among actors but there is limited knowledge regarding which actors dominate conflicts about ES trade-offs and with which power strategies they can dominate these conflicts. Based on insights of the reviewed literature, especially the insights gained by Howe et al. (2014) on trade-off situations, we formulated the following hypotheses to explore in this study:

- (1) Private interests in forest management are more powerful than public interests. Private interests are those interests that are pursued by actors for their individual benefit, for example selling herbs collected in the forest. In contrast, public interests serve the overall society.
- (2) Actors with an interest in the provisioning services of forests are more powerful than actors in regulating and maintenance ESs or cultural ESs.
- (3) Actors acting at the local scale are more powerful in regard to forest management than actors acting at a larger scale.

3. Methods

3.1. Examined countries and case study areas

In contrast to the many existing studies on ESs which take a single case study approach, our study compared the empirical data from nine countries. As Biber et al. (2015) found, forest management practices and the trade-offs between ESs differ widely among European countries. This may reflect the diversity in interests, conflicts and power relations in the different countries. Hence, a comparison of multiple case studies allows for more robust conclusions about the impact of power relations on the provision of ESs. This study is based on empirical data from: Germany,

Table 1

Power evaluation based on the actor-centered power approach (Krott et al., 2014).

Power resource	Definition	Examples
Coercion	Altering behavior with force	Applying real force Threatening with force Bluffing about force Secret non-compliance with rules
Incentives/ Disincentives	Altering behavior with incentives or disincentives	Material incentives, e.g. subsidies Immaterial incentives, e.g. moral arguments
Dominant information	Altering behavior by supplying information that cannot be checked independently by the receiver	Ideology Expert knowledge

Ireland, Italy, Lithuania, Netherlands, Portugal, Slovakia, Sweden, and Turkey. Turkey is not a member of the European Union and only parts of Turkey are geographically located within Europe. Turkey has been included in this study because it is politically, economically, and culturally associated with Europe.

We applied a multi-level case-study design by conducting a general analysis at the national level and an analysis at case study level in each country. In all participating countries, one or two case study areas (CSAs) were defined to analyze the applied forest management practices, to assess forest ESs, and the power relations between actors at a regional level. In the Netherlands, the heavily populated country with small forest coverage, the analysis was conducted for the country as a whole. The other CSAs were:

- Bavaria (West Augsburg) in Germany
- Brandenburg (Lieberose/Schlaubetal) in Germany
- Barony of Moycullen, County Galway in Ireland
- Veneto (Eastern Veneto) in Italy
- Telsiai in Lithuania
- Vale do Sousa in Portugal
- Podpoľanie in Slovakia
- Kronoberg County in Sweden
- Gölcük in Turkey

The included countries and CSAs represented different bio-geographical and socio-economic backgrounds. The forest ownership structure, including the relative weight of public, industrial private, and non-industrial private, differed among countries and CSAs, as did property size (Appendix 1). Due to their diversity of bio-geographical and socio-economic backgrounds, the selected countries and CSAs allowed the inclusion of a wide range of forest management conflicts with different actors involved in ES trade-offs. This diversity of included conflicts and actors allowed a comprehensive testing of our hypotheses regarding which actors dominated conflicts involving ES trade-offs.

3.2. Data collection and analysis

During a four-day seminar in September 2016, the researchers from all nine countries were trained to adhere to a coherent methodological approach in all countries. During this training seminar, the different power strategies of actors were discussed with examples from the countries to train the researcher in conducting an actor analysis. Furthermore, detailed guidelines and examples were developed to ensure coherent analysis.

This study is based on qualitative data from all nine countries. Researchers working in the corresponding countries, with knowledge of the local language and local circumstances compiled the data in each case. We triangulated different sources of information and different qualitative methods of data collection to validate our findings (Flick, 2008). We reviewed documents from different actors (e.g. press releases, position papers, webpages) and conducted desk research to get a deeper understanding of the interests of actors, their power resources, and forest use conflicts. We also conducted 220 qualitative interviews. A list with interview questions (Appendix 2) was developed that guided the interviews. The interviews were conducted in 2016 and 2017. In Ireland, Lithuania, and the Netherlands, interview material from 2012 and 2013, from a former project with similar research focus, was also analyzed.

Interview participants were selected according to the principle of complementarity with the aim to include actors with different interests in ESs, governmental, market, and civil society actors, and actors from different governance levels. For example, environmental, hunting, timber industries and forestry NGOs, forest owners, state forest management and administration, and actors from the recreational sector were included in the interviews. The number of interviews conducted varied from 10 to 40 (Appendix 3). The duration of the interview lasted from 20 minutes to 2 hours, depending on the interview participant. Researchers

recorded the interviews. In some countries, observations, meetings, and discussion with stakeholders within the forests took place. All countries hold two workshops to discuss and present research findings with stakeholders.

The researchers had different foci in their data collection. While in some countries (e.g. Sweden, the Netherlands) data collection relied mostly on documentary analysis, other countries focused on conducting interviews (e.g. Slovakia, Portugal). In some countries only one researcher collected and analyzed the data, in other countries more researchers were involved in the process. This collaborative process of data collection and analysis could have introduced bias. We discussed our results among the researchers of participating countries in order to reduce different interpretations, but we did not establish a systematic process to check for intercoder reliability.

3.3. Actor analysis

First, we identified the actors with interest in forest management in all cases through literature review, document and websites analysis. We included all individual actors and organizations that have an interest in forests, either directly (through involvement in the management and use) or indirectly (by affecting the management) (Freeman, 2010).

Secondly, we determined the interests in ESs of the identified actors. The interests of the actors in ESs were collected through literature review, document and website analysis and in some CSAs also via participatory observations of forest policy processes, e.g. the observation of a forest stakeholder meeting, organized by a governmental agency. Furthermore, we conducted qualitative interviews to verify and extend the information gained from other information sources. The diversity of actors involved in forest management represented a range of interests that are partially conflicting. We identified conflicts derived from ES trade-offs among actors based on the information collected about forest management conflicts. We analyzed the interviews, documents, websites, etc. with the aim to identify forest management conflicts in the examined CSAs and countries. The identified conflicts were compiled in tables for each country, based on the example provided by the data collection guideline. Subsequently, all the conflicts identified from each country were allocated within a single table. Discussions among the researchers took place, if e.g. some of the identified conflicts referred to the same aspect described in different ways, or instead, distinct conflicts were identified.

Finally, we evaluated the power of actors with interest in ESs with the actor-centered power approach (ACP). Based on our data collection, we evaluated the power resources of each actor. The power evaluation is based on the reputation of actors, an analysis of decision-making outcomes of actors in forest management, and an analysis of forest policy processes in each country. For our power evaluation, we triangulated qualitative data from our interviews, document analysis, observations of forest policy processes, meetings with stakeholders and literature review of former studies.

For each actor with an interest in ESs, we evaluated each type of power strategy as to whether its application had a strong (+++), a medium (++), or a low impact (+) on forest management (Hubo and Krott, 2015). The evaluation of power resources takes the relative strength of actors in relation to other actors into account. The total power score is not the average of the capacity to apply the different power strategies. Instead, it represents the overall capacity to impact forest management by applying all power strategies, in relation to other actors. Having strong power resources (+++) means that an actor is able to impact forest management strongly, often against the interests of others. In contrast, low power resources (+) means that an actor can apply the power strategy to some extent but is not able to realize their own interests against the will of others. Having medium power resources (++) implies that an actor can realize interests against the interests of some actors with low power resources, but is not able to impact forest management substantially against the interests of actors possessing

strong power resources. Next, we aggregated the power scores of each power strategy of each actor.

Based on our prior knowledge of forest management, literature review, and document analysis, we defined some focal ESs, which were defined *a priori* to conducting the interviews (e.g. timber production, recreation, water quality, lifecycle maintenance, habitat and gene pool protection). Additional focal ESs were included because they were identified as highly relevant in the analysis of the CSAs (e.g. NWFPs, wildfire regulation). Actors were aggregated into groups, based on their interests into these focal ESs to estimate the overall power resources of all actors supporting a certain ESs. For example, all actors with interests in timber production, recreation, or water quality were aggregated into a group. Thus, for this aggregated power score, which describes how specific ESs are supported by power, we considered the individual strength of actors and the number of actors with specific interests. For example, a single individual with interest in outdoor recreation in forests will usually have only low power resources to influence other actors to realize this interest. In contrast, if large amounts of people use forests for outdoor recreation, and demand for recreational-friendly infrastructure (paths, benches), they can have medium or strong power resources. An individual actor could be aggregated in several groups, if this actor was interested in several focal ESs.

The analysis of power resources evaluated the relative power of actors within one country. Actor's having strong power resources in one country do not necessarily have the same power resources as an actor with strong power resources in another country. We did not define an absolute value of having strong, medium or low power resources, instead we analyzed the relative capacity to apply power strategies.

Finally, we analyzed which actors, or actor groups, could dominate forest management conflicts, making them "winners" or "losers" of such conflicts. We tested three hypotheses about which actors dominate forest conflicts. We assessed for each country, which actors were able to successfully realizing their interests in ESs, and which actors could not realize their interests in conflicts. In many cases, all actors could realize their interest to some extent, thus we assessed which actors could dominate conflicts.

4. Results

4.1. Actors' interests in forest ecosystem services and conflicts about their trade-offs

In all countries and CSAs, the multifunctional role of forests is reflected in a diversity of actors interested in forest management decisions. Actors with interest included governmental actors from different political-administrative levels, the state enterprises that manage public-owned forests, private forest owners (including the organizations and cooperatives representing private owners), pulp, paper and timber industry, and environmental non-governmental organizations (eNGOs). Actors from the tourist sector had interests in forest management in Italy's CSA, whereas in CSAs in Germany, Ireland, the Netherlands, and Slovakia, actors from the water sector were represented, and in the CSAs in Italy and Turkey actors interested in non-wood forest products (NWFPs) (e.g. chestnuts, berries, honey, herbs) were represented. Furthermore, citizens with interest in different types of recreation (e.g. hiking, mountain biking, hunting) are using forests for these activities.

4.1.1. Interest in ESs

Regarding provisioning ESs, there is a strong interest in the provision of wood by public and non-public forest owners and timber processing industries in all CSAs, except in the CSA in Italy. In Italy, domestically produced timber is economically not relevant for the overall economy because of significant timber imports. Forest felling accounting on average for about 29% of the net annual increment at national level, versus an average 66% figure at the European scale (Forest Europe, 2015). Other relevant interests in provisioning ESs in the CSAs included

game, herbs, mushrooms, and chestnut shoots. In all countries and CSAs, hunters had interest in the provision of game. Some income for forest owners is generated by giving out hunting licenses. Different actors (e.g. environmental non-governmental organizations (eNGOs), nature conservation administrations) had strong interests in protecting biodiversity and habitats. The interest of actors in regulating ESs is medium with some variances. In some CSAs, strong interest exists in water quality, wildfire protection, or erosion protection. The interest in cultural ESs is high among the general public, but low among actors with economic interests in forests (except the tourism sector).

4.1.2. ESs conflicts in forest management

In all cases, we identified many conflicts concerning trade-offs among ESs in forest management (Appendix 4). The general trade-offs between provisioning, regulation and maintenance ESs, and cultural ESs cause conflicts in all cases. The scale of conflicts differed. Some of the reported conflicts were local ones identified within the CSAs, other conflicts were identified at a national scale. Many conflicts existed at the local scale but were also relevant at the national scale within political processes and discussions.

Most relevant are the conflicts between the following ESs:

- (1) Timber provision vs. lifecycle maintenance, habitat and gene pool protection (in all countries): Forests managed for optimization of timber production have a lower value as habitats and for biodiversity conservation relative to protected and less intensively managed forests.
- (2) Outdoor recreation and tourism vs. timber provision and lifecycle maintenance, habitat and gene pool protection (in all countries): Local recreationists and tourists use forest for different types of outdoor recreation, object and protest about harvesting measures and desire nice sceneries and landscapes, in some cases also conflicts with nature conservation exist because visitors do not stay on official trails, and disturb wildlife.
- (3) Timber provision vs. provision of high game densities and cultural value of hunting (Germany, Ireland, Italy, Lithuania, Slovakia, Sweden): Hunters have a preference for high game densities to be able to see and shot game. These high game densities have a negative impact on timber production because of browsing and other damage to trees and seedlings.

Conflicts exist between actors with interests in specific competing ESs within the same category of ESs, for example between different types of cultural ESs (e.g. horse riding vs. hunting). Variation in the relative importance of specific conflicts exists between local, regional, and national levels. For example, one of the main sources of conflict for forest owners and managers at a local level is with groups of (individual) visitors. Examples include forest visitors not agreeing with management activities (e.g. harvesting, cutting of (monumental) trees or removal of forest for conserving open landscapes such as heath land), and forest visitors not following the rules (e.g. not having dogs on a leash, mountain biking in areas where it is not allowed). This type of conflict is not considered relevant to the political process at national levels. Some conflicts are more country-specific because they relate to specific biogeophysical conditions (e.g. increased occurrence of wildfires in Portugal). Other conflicts relate to specific cultural conditions, for example, the conflict between timber production vs. the availability of the forage consumed by Sami reindeer populations in northern Sweden.

4.2. Power relations in forest management

Differences exist in the relative power resources among actors with interests in different ESs in the ten CSAs (Table 2).

Overall, the most supported ESs among the countries assessed were timber and lifecycle maintenance, habitat and gene pool protection, followed by game, water quality, and carbon sequestration. Some

Table 2

Various ES and the extent to which these are supported by power at a regional level.

Case Study Areas	West-Augsburg (Germany)	Lieberose/Schlaubetal (Germany)	Barony of Moycullen (Ireland)	East Venet (Italy)	Telsiai (Lithuania)	Netherlands (country assessed as a whole)	Vale do Sousa (Portugal)	Podpolanie (Slovakia)	Kronoberg County (Sweden)	Gölcük (Turkey)
<i>Provisioning services</i>										
Wood	+++	+++	+++	++	+++	+++	+++	+++	+++	+++
Game	++	++	++	++	++	++	+	++	++	+
Medical plants/ Herbs/ Mushrooms/ other NWFP	+	+	+	++	+	+	+	+	+	+
<i>Regulation and maintenance services</i>										
Water quality	++	++	+++	+	+	+++	+	+++	++	++
Lifecycle maintenance, habitat and gene pool protection	+++	+++	+++	+++	+++	+++	+	+++	++	+++
<i>Cultural services</i>										
Outdoor recreation	++	+	+	+	++	+++	++	+	+	+
Aesthetic values	++	++	+	++	++	++	+	+	+	+
Tourism	+	++	+	+++	++	+++	+	++	+	++

Legend: + (Low), ++ (Medium), +++ (Strong).

countries stood out from this general picture: In Portugal, cork was an important ES, though less relevant in the Portuguese CSA. In the Netherlands actors with interests in recreation had strong power resources.

4.2.1. Provisioning services

In the CSAs of Sweden and Portugal, actors from forestry and timber industries with prior interest in *timber and pulp provision* had very strong power resources to impact forest management. Sweden is the world's third largest exporter of forest products (SFIF, 2018) and the forest sector accounts for approximately 3% of GDP (Forest Europe, 2015, p. 182). Industrial round wood is supplied by forest owned by state and private companies, as well as small-scale forest owners, the latter group owning half of the productive forestland at the national level. The importance of the Portuguese timber and pulp industry is highlighted by its contribution to 7.4% of national exports (DGAE, 2017). The area of eucalyptus expanded substantially to reach 26% of the forest land (ICNF, 2019a), driven by demand from the industry. The latter owns about 5.20% of the forest land (AIFF, 2013) and sets financial incentives to increase eucalypt supply from non-industrial private owners. Its impact on forestry derives further from its investment in nurseries, outreach and research targeting the enhancement of management practices by forest owners in the CSA Vale do Sousa. The wood processing industry also had a strong impact on forest management in other countries, e.g. in the state forestry of Slovakia via a political line.

In the Italian CSA, the main NWFPs are *truffles*. Although the area does not offer top-quality truffles, still these NWFPs can represent an important resource for landowners in terms of income differentiation and integration. Collection and marketing, however, still remain largely informal, although a recent fiscal-regime reform has helped the emergence of these "grey" activities and associated revenues. Additional NWFPs, marginally produced in the Italian CSA, included *pine nuts* - collected by a local cooperative, with an estimated value of about 200–300 €/ha/year - and *wild asparagus*. The collection of some NWFPs (e.g. mushrooms, truffles, chestnuts, cork, pine nuts) also assumed great importance as a recreational activity (i.e. as a cultural rather than just a provisioning service), also generating revenues for local economies (restaurants, hotels, museums, etc.). In the Turkish CSA, chestnut shoots and mushrooms were two important NWFPs. Two to three years Chestnut shoots were used for handmade chairs and the state forest

enterprise was unable to supply the demand. Thus, the sector either imported the required material from China or ordered it from private forest owners at high costs. A "Chestnut festival" has been organized by local people for the last 13 years and celebrated by nearly 1000 inhabitants each year. Thus, actors with interest in NWFPs created material incentives for forest owners to support the provision of NWFPs.

The interest in *energy provision* became increasingly important for forest management in several countries (Germany, Ireland, Italy, Lithuania, Portugal, Slovakia, Sweden, Turkey), because of the construction of dams, wind turbines, or solar farms within forests and the increased demand for biomass for energy production. Actors from the energy sectors became increasingly powerful in forest management during the last decade in all countries. Actors from the bioenergy sector provided high financial incentives for the use of forest biomass for energy. For example, in Slovakia, very high demand for biomass exists, and due to limited wood processing factories with higher added value production, lots of wood is sold to biomass burning plants or is exported. Although this provided income for forest owners on the one hand, on other hand it also resulted in a loss in quality wood, which was not well appreciated on the market. This development was also observed in other countries (e.g. Germany, Italy).

Actors interested in hunting *game* were indirectly powerful in most cases because of the importance of browsing damage to forest management (Germany, Lithuania, Slovakia, Sweden). For example, in Lithuania and Sweden, regenerating pine has always been problematic due to browsing. Fencing is extremely expensive and is used mostly to protect only the most valuable plantations, primarily oak. Thus, inefficient game management has led to prioritizing spruce over other tree species. The common opinion of forestry professionals is that "lead is the best repellent". On the other hand, hunting can be considered as a part of, for example, Lithuanian culture, and the hunters' associations had quite a strong influence on state forest policy due to their financial resources, and the often close personal relations with decision-makers. The situation in Slovakia is similar, where hunting used to be a common activity, but has since turned into more of a hobby of the rich and powerful. Hunters' clubs have been suggested by policy makers, media and (up-to-some extent) public to be structure for "under-the-table settling of issues" between decision-makers and foresters. It is important to note that many foresters and forest policy decision-makers in Slovakia used to have a big interest in hunting themselves, which made the

hunting lobby even more powerful.

4.2.2. Regulation and maintenance services

The group of actors interested in these ESs combines governmental, civil society, and some market actors, e.g. the (eco) tourism sector. This coalition of different actors allows interest in *lifecycle maintenance*, *habitat and gene pool protection* to be supported by actors with strong power resources in all CSAs except Portugal and Sweden. Governmental actors from nature conservation authorities apply coercion by implementing nature conservation laws and provide incentives by paying subsidies to private forest owners for forest management that has a high value for biodiversity and habitats. In some countries, incentives were the most important power strategy of governmental actors (e.g. the Netherlands). In other countries (e.g. Slovakia, Lithuania) coercion was more important to steer private forest owners, though financial support was also provided for nature protection to non-state forestry in Slovakia. In Slovakia, actors interested in biodiversity and nature protection (governmental actors and eNGOs) had very strong power resources. For instance, in nature protected areas they had a heavy influence in forest management plan elaboration and implementation.

The forest certification organizations FSC and PEFC provided incentives for public and private forest owners towards forest management that complied with their standards, including considering ecological aspects in forest management. Nationwide active eNGOs influenced forest policy processes by lobbying and organizing large campaigns in favor of strengthening supporting ESs in forest management. These eNGOs provided immaterial incentives by appealing to the welfare of future generations or the moral responsibility for biodiversity conservation.

Some market actors, such as the tourism sector in the CSA in Italy and the CSA Germany, Lieberose/Schlaubetal had interest in forests that have a high biodiversity value and are habitat for rare species (e.g. wolves) of interest to forest visitors. Also large parts of the general public that are using forests for recreation have preferences for close-to-nature forests, as long as forest access is not restricted. However, some parts of the public was also threatening to the maintenance of supporting ESs, by illegal littering. For example, in the Netherlands, illegal dumping took place in forests of barrels with the oil or chemicals byproducts of illegal drug production.

The power resources behind regulating ESs were medium in most CSAs. In Ireland, the Netherlands, and Slovakia actors interested in *water quality* had strong power resources. Water quality concerns in Irish forestry mainly related to habitat quality for the freshwater pearl mussel (*Margaritifera margaritifera* L.), which are negatively impacted by siltation, acidification, and eutrophication caused by forestry operations (Forest Service, 2000). Ireland is home to 46% of the EU Freshwater Pearl Mussel individuals (Forest Service, 2008) and Natura 2000 designations are used to protect freshwater pearl mussel habitats, limiting the ability to intensively manage forests in and near these areas. In 2006, a harvesting moratorium was placed on the forest in one of the priority freshwater pearl mussel catchments to mitigate any adverse environmental impacts from forestry operations. In the Netherlands and Slovakia, water companies that extract drinking water from forests, generally without compensating landowners, had strong power resources.

In Portugal, the power resources of actors interested in regulating ESs (e.g. *wildfire protection*) increased substantially as a consequence of the 2017 wildfire season. This was a year characterized by extreme wildfire events. The burned area extended over 537,000 ha (ICNF, 2019b) and the loss of human lives was unprecedented (116 victims). Because of these catastrophic events, the Portuguese government revised forest legislation, namely to restrict the afforestation and reforestation with eucalypt plantations, and to set stricter criteria for the management of forest biomass as fuel for wildfires and to enforce its implementation (e.g. by imposing fines on non-compliant landowners).

4.2.3. Cultural services

In some cases, economic interests in ESs other than timber had strong power resources, for example in the CSA in Italy, actors from the tourism sector with strong interest in cultural ESs had strong power resources. The Italian CSA, which consisted of a mosaic of forests, was located within a highly populated area (about 140,000 inhabitants) and attracted a huge number of tourists and visitors. Being located close to Venice and other historical sites and beach places along the Northern Adriatic Sea, the area was visited every summer by about 3 million of visitors, both locals and from Central, Northern and Eastern Europe, who demanded green places for hiking as well as for shelter. Some of these areas were already equipped with certain infrastructure, like hiking or bike paths, and were more and more used by local organizations to offer forest-based activities and services, including for example horse trails, environmental education activities and sport events. Also in the Lithuanian CSA, tourism and outdoor recreation were important, though the amount of visitors was not comparable to the CSA in Italy.

The groups of actors with prior interests in *outdoor recreation* mainly consisted of actors with low power resources, except in the Netherlands. As a heavily populated country with relatively few forests (~10% of the land surface), the recreational value of forests contributes substantially to the national economy in the Netherlands. The leisure sector benefits most from the forest presence. This goes together with increasing societal demand for nature and natural resources due to a growing need for wellbeing that is associated with more green areas and the trend of green care initiatives (e.g. forest bathing). However, forest owners were not compensated (much) for the use of their forests, unless they included leisure as part of their economic management model, like renting recreational houses, exploiting a golf course, a restaurant and other forest related facilities.

4.3. Actors dominating conflicts about forest ESs

We examined three hypotheses about which actors dominated in conflicts about forest ESs in the CSAs (Table 3).

(1) Private vs. public interests

We examined the hypothesis that private interests in forest

Table 3
Overview about hypotheses testing in all CSAs.

	Private interests dominate public interests	Provisioning interests dominate broad range of interests	Local interests dominate national interests
Germany (West Augsburg)	–	–	+
Germany (Lieberose/ Schlaubetal)	–	–	+
Ireland (Barony of Moycullen)	–	–	–
Italy (East Veneto)	–	–	+
Lithuania (Telsiai)	–	–	–
The Netherlands (The Netherlands)	–	–	+
Portugal (Vale do Sousa)	+	+	+
Slovakia (Podpolanie)	–	–	–
Sweden (Kronoberg Country)	+	+	+
Turkey (Gölcük)	–	–	–

Legend: + (Hypothesis confirmed), – (Hypothesis not confirmed).

management are more powerful than public interests. This hypothesis was only supported in Sweden's and Portugal's CSA, in which interests in forestry profits, combined with an industrial interest in round wood, dominated forest management. In contrast, in all other countries, public interests dominated or had comparable power resources to actors with private interest. Actors with interest in lifecycle maintenance, habitat and gene pool protection sometimes dominated in forest management conflicts. For example, in the CSA in Italy, the tourism sector was very powerful with its private interest in using cultural ESs for generating income from tourism but actors with interest in lifecycle maintenance, habitat and gene pool protection ESs had comparable power resources.

(2) Provisioning vs. broad range of interests

We examined the hypothesis that actors with an interest in the provisioning services of forests are more powerful. This hypothesis was supported in Sweden's and Portugal's CSAs. In Sweden the interests in wood and game provision were supported by strong power resources. In Portugal, only the interest in wood provision was supported by strong power resources. In all other countries, the actors interested in lifecycle maintenance, habitat and gene pool protection also had strong power resources. In the CSA in Italy, actors dominating conflicts had an interest in cultural ESs because of the importance of tourism and recreation for the local economy. In the Netherlands, actors with an interest in cultural forest ESs were winners in ES conflicts because of the relatively low (but growing) economic importance of forestry and the low share of forestland, resulting in a high user pressure by the outdoor recreation sector. In Ireland, the actors interested in water quality were very powerful because of the occurrence of the rare freshwater pearl mussel. In Portugal, actors with interest in fire regulation dominated forest management conflicts because of severe wildfire events. Thus, there were many examples from our cases that contradicted this hypothesis.

(3) Local vs. national interests

We examined the hypothesis that actors from the local scale were more powerful in regard to forest management. This hypothesis was supported in some CSAs because many actors (e.g. citizens, companies) directly impacting forest management were active at a local scale. We identified three different factors why local interests often dominated forest management conflicts:

1. *Lack of law enforcement at a local level:* National level actors (e.g. forestry ministries and administration) are often theoretically powerful but not in forest management practice because laws were not enforced on the ground (e.g. in Portugal).
2. *Local actors disobey laws:* Furthermore, actors at a local scale were powerful by realizing their interests by disobeying laws. For example, dumping garbage in the forest is legally forbidden in all cases but nevertheless occurs. It is impossible to fully observe the forest at all the time to avoid illegal activities. This lack of ability of forest owners and public administration leads to illegal hunting, infringements of nature conservation laws, or illegal berry or mushroom picking by local actors interested in cultural and provisioning ESs.
3. *Local actors know their forest:* The power strategy dominant information could be applied by local actors because they had the best knowledge of local conditions. For non-local actors it was often difficult to check the information about forests provided by local actors. However, differences existed among countries. In cases with mandatory and/or centralized forest management planning (Lithuania, Slovakia, Turkey) the forests were very well monitored. In countries without centralized forest management planning, forestry inventories and species monitoring were done but their level of detail could not compete with local knowledge about certain forests.

However, we also identified four factors, which allow national actors to dominate forest management conflicts:

1. *Nationwide mandatory forest management plans:* In Slovakia, certified forest managers, who forest owners have to pay by themselves, applied mandatory forest management plans. Though, mandatory forest management plans are elaborated in participatory way. Implementation of forest management plans was also overseen by forest and nature protection state administrators. Therefore, state actors were more powerful in Lithuania and Slovakia than in countries where private forest owners had either stronger property rights (e.g. in Sweden or the Netherlands), or where less control of forest management took place (e.g. in Portugal).
2. *Centralized state forest service:* In Turkey, the state forestry service was organized centrally, with officials setting up relevant plans implemented in all state forestry sub-divisions. Although local state forest enterprises made decisions and used initiatives, generally local communities/NGOs did not have enough power or had no awareness of forestry issues. Yet, forest enterprises in Turkey were not isolated from the society, if requests were acceptable, they were generally taken into account. In addition, centralized organization and planning process in Turkey followed nationwide interests via strategic and tactical plans, primarily prepared for by the planning department at the headquarters of the General Directorate of Forestry. Those plans must be prepared based on management guidelines set by the department which were, in fact, set by legal regulations.
3. *EU-Regulations:* In Ireland the local interest in timber harvesting was dominated by the European interests to protect the Freshwater Pearl Mussel (though also local actors also had an interest in biodiversity conservation). Also, the use of biomass as a renewable energy source promoted by the EU lead to increased pressure for biomass use in all countries.
4. *Nationwide discourses:* The power of nationwide NGOs and lobbying is based on the provision of immaterial incentives in discourses, by appealing to social conventions, morals, and the welfare of future generations (e.g. avoid species extinctions).

5. Discussion

5.1. Discussion of results

Our mapping identified a high variety of actors, forest use conflicts, and power relations across the investigated countries. However, there were also similarities in all the nine countries and ten CSAs examined. Because human intervention may support the provision of some ESs, with a reduction in the availability of other ESs, decision-making about forest management means fulfilling the interests of some actors, at the expense of others. Howe et al. (2014) state that there is a lack of studies examining ESs trade-offs and synergies in cultural and regulation and maintenance services. In our study, we showed that cultural and regulation and maintenance ESs play an important role in conflicts about forest ES trade-offs. Especially in areas with high population density, interest in forest recreation creates conflicts with other interests in forest ESs. Actors interested in provisioning ESs have strong power resources, but because of specific bio-geophysical, political or economic conditions, actors with interest in regulation and maintenance ESs or cultural ESs can have equal or stronger power resources. Catastrophic events, such as large-scale wildfires or the high relevance of climate change effects (e.g. wind, fire, ice, bark beetle) in the political discourse opened up political windows of opportunity to make actors with interest in non-provisioning ESs very powerful in forest management.

We tested three hypotheses about which actors are able to dominate conflicts about ES trade-offs. We compared the power resources of (1) actors with private vs. actors with public interests, (2) actors with interest in provisioning ES vs. actors with a broad range of interests, and (3) local actors vs. national actors. However, there are many examples

for actors who cannot clearly be assigned to one of these categories. For instance, the private and public interests were closely linked to each other. For example, effective wildfire regulation was a public interest in Portugal but also served the private interests of forest owners and pulp companies. Cultural ESs serving tourism (as well as other ESs associated to them, e.g. carbon sequestration and biodiversity enhancement) are public goods, benefiting the whole society. Thus, in several cases, it was difficult to distinguish clearly between private and public interests in ESs. The private/public interest dichotomy can vary, in terms of intensity, also depending on the ESs considered and in regard to what kind of human activities were actually performed in the field. Furthermore, the allocation of private property rights in forests affected conflicts among private and public interests. For example, in Turkey (almost) all forests are owned by the state, consequently the interest in economic profitable management of forests is also a public interest. Furthermore, some actors act at local, regional and national scale. Thus, in some identified conflicts it was difficult to say if an involved actor was a “local actor”.

Howe et al. (2014) show that actors who were losers of ES trade-offs were not evenly distributed among different categories of actors, with 64% being local actors, 47% regional actors, 19% national actors and 29% global actors. This implies that not only were the majority of winners, but also the majority of losers of trade-offs act at a local scale. Our findings in this study about the dominance of local or national actors in conflicts are also mixed. Therefore, we suggest further research focusing on the question of whether the scale of action has any systematic impact on interest realization in forest ES conflict.

Some studies (e.g. Felipe-Lucia et al., 2018) see the outcome of forest ES bundles only as product of forest attributes (e.g. DBH, vertical heterogeneity and shrub richness) and environmental factors (e.g. slope, soil depth). However, the interests of actors and their power resources also have an impact on ES bundles. Schaich and Plieninger (2013) show that forest ownership has an impact on stand structure and ES bundles, e.g. the amount of carbon sequestration. Therefore, the “human factor” should not be completely neglected in studies about forest ESs. Our examined CSAs differed in their ownership types and sizes. Property rights are an important factor determining possible power strategies and access to ESs. Our study focused on power relations in specific ES trade-off situations but did not cover the complex power relations of all actors interested in forest management. Every actor stands in multiple social relations, so called ‘webs of power’ to other users by gaining, controlling and maintaining access to ESs (Ribot and Peluso, 2003). Actor A might be a “winner” in a conflict with actor B about one trade-off and a “loser” in a conflict about another ES with actor C. A systematic mapping of ‘webs of power’ in ES assessments could help to incorporate power relations in forest and environmental management. Knowledge about power relations in ES trade-offs is a necessary first step to deal with them. ‘Webs of power’ can be changed, for example, by changing the allocation of property rights related to forest ecosystems.

5.2. Discussion of methods

The data provided by this study was collected and analyzed within a large, international research project with more than 30 researchers from nine different countries. This allowed us to include detailed information from each country, but the necessary collaboration of an international team to do so also presented challenges in terms of organizing the research. The interpretation of results may have been biased in different ways by the individual perceptions of researchers and the different methods applied for data collection. We discussed and compared our results among the researchers of participating countries in order to reduce different interpretations, but we did not establish a systematic process to check for intercoder reliability or bias check. We recommend that future research projects develop and plan for such a process from the first stage of project development, or in the early project planning phase. We met for a data collection workshop at the beginning of the

project in person but it would have been very helpful to also have a data analysis and interpretation workshop together with all involved researchers and/or to establish a systematic intercoder reliability check among countries.

The power strategies were evaluated within countries, so no exact calibration contrasting absolute power resources among countries took place. Therefore, the presented capacities to apply power strategies display power relations detailed within countries only. The analysis of power strategies applied within the CSAs was based on specific examples from forest management on the ground. In contrast, the analysis of power strategies at a national level provided a more general picture of the examined countries.

The applied analytical framework (ACP) by Krott et al. (2014), which evaluates three different power strategies: coercion, (dis)incentives, and dominant information. A refinement of power strategies within the ACP approach could enhance power analysis further. Specifically, the category “coercion” included many different aspects of exercising power, such as organized coercion by governmental actors that are implementing laws, illegal activities by the general public, or the property rights of private forest owners. A stronger systematization of the different aspects involved in having the capacity for coercion would strengthen the analytical value of power analysis. Furthermore, the power strategies material and immaterial incentives differed widely in their function. Separating these into two distinct power categories could be a reasonable path forwards. Thus, we suggest a modification of the ACP framework (Table 4). Future research will be needed to test applicability and usefulness of our suggestions.

We followed the suggestion of Hubo and Krott (2015) to evaluate if an actor has strong (+++), medium (++), or low (+) power resources to impact forest management. Obviously, there are more than three levels of power resources, implying that there can still be differences in having medium power resources. Therefore, we suggest for future studies applying the ACP approach to do a rating of power resources along a continuum, instead of differentiating into three categories of power resources.

Table 4
Suggestions for modifying the ACP framework.

Power resource	Definition	Sub-categories	Examples
Coercion	Altering behavior with force	Organized coercion	Governmental actors that are implementing laws.
		Property rights	Implementing (forest) ownership rights.
		Threatening with force	Forest owner threatening to fence his/her forest.
Material Incentives/Disincentives	Altering behavior with material incentives or disincentives	Bluffing about force	Governmental actors announce to implement laws, without having the actual capacity to do it.
		Non-compliance with laws	Illegal activities (e.g. waste disposal, illegal logging).
			Taxes or tax reductions forest management concepts.
Moral appeal	Altering behavior by providing immaterial incentives by e.g. appealing to moral or norms		Subsidies for allowing nature conservation measures within a private forest.
Dominant information	Altering behavior by supplying information that cannot be checked independently by the receiver		Media campaign for nature conservation.
			Ideology, e.g. about traditional forest management.
			Expert knowledge, e.g. about the occurrence of a certain endangered species.

Another limitation of our study concerns the potential for different understanding of concepts and terms. Interviewees were not systematically asked how they would define different ESs, thus different interpretations of some ESs may also have influenced outcomes. For example, the concept of biodiversity and the related ES “Lifecycle maintenance, habitat and gene pool protection” was very broad. Biodiversity could have differing definitions across countries and across different types of actors (e.g. recreationists, conservationists, or foresters). Dietze et al. (2018) showed that the understanding of the term “ecosystem services” can differ widely among different farmers and between farmers and scientists. We recommend for future research projects to ask interviewees how they understand and define key concepts.

Another limitation of our study is the overrepresentation of interests of powerful actors in our results. Due to our methods of data collection and our focus, we focused on those actors who were powerful in forest management and visible in forest policy discourses. Our research focus largely neglected the interests of powerless actors who were not able to realize their interests in ESs in forest management.

6. Conclusions

Many studies describe conflicts among actors but there is limited knowledge regarding which actors dominate conflicts about ES trade-offs and with which power strategies they use to dominate these conflicts. With this study, we contributed to filling this research gap. In our analysis, we distinguished between power based on coercion, (dis)incentives, and dominant information based on the actor-centered power approach by Krott et al. (2014). Our analysis showed that the capacity of different actor types to apply these different power strategies varies. Governmental actors have more options to realize their interests based on coercion. Many market actors provided material incentives, and civil society actors were successful at providing immaterial incentives by making moral arguments and emphasizing the welfare of future generations in their public discourses and the political process. We suggested some modifications of the actor-centered power approach to enhance the analytical clarity of power analysis. Future research is needed to enhance the knowledge of the role of power and relations among actors in the generation of ES bundles of forests and of other ecosystems.

Our study showed the value of examining power relations of actors involved in ES conflicts. A too technocratic perspective on the optimization of ESs would not lead necessarily to successful implementation of new, innovative forest management concepts that produce win-wins because of the interests and power relations of those actors that drive forest management. We agree with the suggestion by Howe et al. (2014) that the ES assessment is best understood as a framework for analyzing relationships between society and nature, rather than as a decision-making tool, because forest use and the resulting ES bundles are an expression of power relations among actors rather than the result of a technocratic deliberation process. Theoretical win-wins are often not possible to implement in practice if they are detrimental to the interests of powerful actors. Therefore, an analysis of power relations of actors with interest in ESs needs to be included in ES assessments.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoser.2021.101290>.

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